Implementation of Client Scripts & UI Policies

SmartBridge Project Detailed Documentation & Implementation Guide

SERVICENOW DEVELOPMENT

DOMAIN / PLATFORM

ServiceNow System Administration & Development

PRIMARY OBJECTIVE

Dynamic Form Control & Client-Side Validation

📌 Project Overview & Purpose

This project focuses on automating form behavior, enforcing data validation standards, enhancing User Experience (UX), and dynamically controlling field properties in ServiceNow without requiring full server-side processing.

Phase-Wise Project Breakdown

Phase 1: Brainstorming & Ideation Phase

- **Problem Statement:** Standard forms often suffer from user data entry errors, missing critical details, or overwhelming UI complexity when all fields are visible simultaneously.
- **Proposed Solution:** Implement **Client Scripts** (for real-time validations, field updates, and alerts) and **UI Policies** (for no-code field visibility, mandatory status, and read-only toggles).
- **Target Audience:** IT Support Agents, System Administrators, and End Users interacting with Incident/Change records.

Phase 2: Requirement Analysis Phase

- **Functional Requirements:**
  - **UI Policy 1:** Set `Resolution Notes` and `Resolution Code` mandatory and visible only when State = *Resolved*.
  - **UI Policy 2:** Set high-priority incidents (Priority = 1) to Read-Only once closed.
  - **Client Script 1 (onLoad):** Display a warning banner for critical (Priority 1) tickets.
  - **Client Script 2 (onChange):** Validate that the contact number contains exactly 10 digits.
  - **Client Script 3 (onSubmit):** Block form submission if description length is less than 20 characters.
- **Technical Requirements:** ServiceNow Developer Instance (PDI), JavaScript, ServiceNow Client API (`g_form`).

Phase 3: Project Design Phase

> **Execution Workflow:** Form Load → `onLoad` Script & UI Policies Trigger → Field Input → `onChange` Script & UI Policy Conditions → Form Submit → `onSubmit` Script Validation → Save/Block Execution.

Feature	UI Policy	Client Script
Primary Purpose	Show/Hide, Mandatory, Read-Only fields	Complex validation, string logic, custom alerts
Code Requirement	No-Code / Low-Code	JavaScript (`g_form` API)
Execution Timing	On Load & Field Change	`onLoad` , `onChange` , `onSubmit` , `onCellEdit`

Phase 4: Project Planning Phase

Recommended task distribution for a 4-member team structure:

- **Member 1 (Team Lead):** ServiceNow Instance Configuration, Table Selection, Phase 1 & 2 Docs.
- **Member 2:** Implementation of Client Scripts (`onLoad` , `onChange` , `onSubmit`).
- **Member 3:** Implementation of UI Policies and UI Policy Actions.
- **Member 4:** Testing, GitHub Repository Management, and Demo Video Preparation.

Phase 5: Project Development (Code & Configuration)

1. UI Policy Configuration:

- **Table:** Incident [`incident`]
- **Condition:** `State IS Resolved`
- **Actions:**
  - `close_notes` → Mandatory: **True**, Visible: **True**
  - `close_code` → Mandatory: **True**, Visible: **True**

2. Client Script Implementations (JavaScript):

A. onLoad Script (High Priority Warning Banner):

```
function onLoad() {
    var priority = g_form.getValue('priority');
    if (priority == '1') {
        g_form.addErrorMessage('CRITICAL INCIDENT: Please handle with high priority!');
    }
}
```

B. onChange Script (10-Digit Phone Validation):

```
function onChange(control, oldValue, newValue, isLoading, isTemplate) {
    if (isLoading || newValue === '') {
        return;
    }
    var phonePattern = /^\d{10}$/;
    if (!phonePattern.test(newValue)) {
        g_form.showFieldMsg('u_contact_number', 'Please enter a valid 10-digit phone number.', 'er
        g_form.setValue('u_contact_number', '');
    }
}
```

C. onSubmit Script (Minimum Description Length Check):

```
function onSubmit() {
    var description = g_form.getValue('description');
    if (description.length < 20) {
        alert('Description must be at least 20 characters long.');
        return false; // Prevents form submission
    }
}
```

Phase 6: Project Testing Phase

Test Case ID	Scenario	Expected Outcome	Status
TC_01	Change Incident State to 'Resolved'	Resolution Notes & Code become visible & mandatory	PASS
TC_02	Enter "12345" in Contact Field	Field error message appears; value cleared	PASS
TC_03	Submit form with short description (<20 chars)	Alert displays and form submission is halted	PASS

Phase 7: Project Documentation Phase

Repository Folder Structure for GitHub submission:

```
├── Phase_1_Ideation/
├── Phase_2_Requirements/
├── Phase_3_Design/
├── Phase_4_Planning/
├── Phase_5_Development/
│   ├── Client_Scripts/
│   └── UI_Policies/
├── Phase_6_Testing/
├── Phase_7_Documentation/
└── README.md
```

Phase 8: Project Demonstration Phase

> **Video Script Structure:**
>
> 1. **Introduction:** State Project Name, Team Members, and Roles.
> 2. **Purpose:** Explain how Client Scripts & UI Policies improve form validation and UX.
> 3. **Live Execution:** Show `onLoad` warning, `onChange` phone validation error, `onSubmit` short text blocking, and dynamic `UI Policy` field appearance.
> 4. **Output:** Complete successful submission and wrap up with credits.